

Gatekeeper Model v2: Epigenome-Transcriptome Integration

S. coelicolor A3(2) M145 | 3 Timepoints | 9 Hypotheses Integrated

Flow

LAYER 1

LANDSCAPE REMODELING

4mC-expression
correlation

$p = 0.139$
perm $p = 0.002$

MTase expression
change

SC_RS17645 ↓
→ 260 sites lost

CCGG/GCCGGC
turnover

Jaccard = 0.000
Complete positional shift

LAYER 2

PROTECTION / DEPLETION

TF binding sites
depleted

fold = 0.66
 $p = 6.8 \times 10^{-5}$

σ factor -10 box
depleted

$p = 1.73 \times 10^{-12}$

Literature 37 TFs
unmethylated

92% methyl-free
SARP 100%

DMGs: no
functional enrich.

Position-dependent
Function-independent

LAYER 3

SIGNAL GATING (Revised)

62 non-literature
regulators

All outside 37 list
15.6% of 1,055 methylated

TCS asymmetric
methylation

7 pairs identified
SK methylated / RR free

T3 arm enrichment
(gain + up)

OR = 8.05, $p = 0.001$
87% at arms

COG T enriched
(signal transd.)

OR = 2.47, $p = 0.045$
COG Q = 0

EXCLUDED: Methylation → TF cascade → BGC activation

H4: 0/37 literature TFs show methylation-expression coordination | SARP activators (redD, actII-ORF4, cdaR, cpkO) ALL unmethylated